A Major Project Synopsis

On

Project ID - BTP2CSE105

PerfTrack AI

Online ML- Based Academic Performance Analyser

Submitted in partial fulfilment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

Session 2025-26 in

Computer Science and Engineering

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March, 2026

Introduction

The landscape of modern education is undergoing a seismic shift, transitioning from traditional pedagogical methods to a sophisticated, data-driven ecosystem. As educational institutions digitize their operations, they generate an unprecedented volume of granular data—ranging from daily attendance and Learning Management System (LMS) engagement to internal assessments and final examination results. However, despite the abundance of this "big data" in education, a significant gap remains: the transition from **descriptive** data (understanding what happened) to **predictive** insights (understanding what will happen).

In the current status quo, evaluation systems are primarily retrospective. They act as a post-mortem of student performance, identifying academic deficiencies only after the final grades are published. This reactive approach is inherently flawed; by the time an educator identifies a struggling student, the window for effective intervention has often closed. This delay not only leads to higher attrition and failure rates but also fails to provide students with the personalized support required to navigate complex curricula.

The **Online ML-Based Academic Performance Analyzer** is designed to dismantle this reactive paradigm. By integrating historical datasets with real-time academic indicators, the system employs advanced Machine Learning (ML) techniques to provide a proactive "early warning" framework. Rather than waiting for a semester-end failure, the system analyzes patterns in attendance, assignment consistency, and quiz performance to forecast future outcomes with high precision. This foresight allows for a shift in the educational philosophy—from judging results to nurturing progress.

Central to this technological leap is the application of supervised learning algorithms. By leveraging models such as **Logistic Regression** for probability estimation, **Random Forest** for handling complex, non-linear variables, and **Support Vector Machines (SVM)** for high-dimensional classification, the system can categorize students based on their risk profiles. These algorithms do more than just predict grades; they provide actionable intelligence that uncovers the underlying drivers of student success or struggle.

Ultimately, the goal of this system is to empower the entire academic community. For educators, it serves as a diagnostic tool to prioritize resources for at-risk students. For students, it offers a transparent view of their academic trajectory, encouraging self-regulation and timely improvement. By turning raw data into a strategic asset, the Online ML-Based Academic Performance Analyzer fosters an environment where every student has the opportunity to succeed before the final bell rings.

Problem Statement

In the modern education system, institutions continuously generate large volumes of student-related data through attendance records, internal assessments, assignments, examinations, and online learning platforms. However, despite the availability of such rich datasets, most institutions still rely on traditional evaluation methods that primarily focus on final examination results. These conventional approaches provide only a retrospective analysis of student performance and fail to identify learning gaps or struggling students at an early stage. As a result, timely intervention becomes difficult, which can negatively impact academic outcomes and overall student development.

Another significant challenge is the inability of existing systems to consider multiple influencing factors simultaneously. Student performance is not solely dependent on exam scores but is also influenced by various parameters such as study habits, attendance patterns, engagement levels, socio-economic background, and participation in extracurricular activities. Traditional systems are not capable of analyzing these diverse factors collectively, leading to incomplete and less accurate assessments. Furthermore, the lack of real-time monitoring and predictive capabilities limits the effectiveness of decision-making by educators and academic administrators.

With the increasing adoption of digital learning environments and online education platforms, the volume, variety, and complexity of educational data have grown significantly. This creates a need for intelligent systems that can process large datasets efficiently, extract meaningful patterns, and provide actionable insights. Machine learning techniques offer a promising solution to this problem by enabling automated analysis, pattern recognition, and prediction of student performance based on historical and real-time data.

Therefore, there is a strong need to develop an advanced, machine learning-based academic performance analyzer that can accurately predict student outcomes, identify at-risk students at an early stage, and assist educators in taking proactive measures. Such a system will not only improve academic performance but also support data-driven decision-making, enhance teaching strategies, and contribute to the overall effectiveness of the education system.

Literature Review

Table 1:

Authors and year	Models / Techniques Used	Data set attributes used	Outcome	Key Findings
Sharmistha Ghosh et al. (2025)	Decision Tree, Random Forest, XGBoost	Academic data including marks, attendance, extracurricular and behavioral features	Achieved up to ~95% accuracy with XGBoost	XGBoost performed best; feature engineering and dashboards improved analysis
Wesam Ahmed et al. (2025)	Linear Regression, KNN, XGBoost, AdaBoost, Random Forest, Voting Regression, SHAP, LIME	Two datasets (10,000 & 6,607 samples) including study habits, attendance, demographics	Voting Regression achieved highest performance (R^2 ~0.989)	Ensemble models improve accuracy; explainable AI enhances interpretability
Meng Wu et al. (2025)	Decision Tree, Naïve Bayes, AdaBoost	Student behavior data (assignments, video activity, grades)	AdaBoost achieved ~73% accuracy	Behavioral data strongly influences performance; AdaBoost outperforms others
Nitin R. Yadav & Sonal S. Deshmukh (2023)	ANN, SVM, Naïve Bayes, Decision Tree, Linear Regression	Educational datasets (attendance, marks, demographics, learning behavior)	SVM achieved highest accuracy (~88%)	Feature selection is critical; ML improves prediction significantly
Ahmed et al. (2024)	SVM, Random Forest, Decision Tree, Naïve Bayes, KNN	Academic data including grades and study behavior	SVM achieved highest accuracy (~96%)	Parameter tuning improves accuracy; ML effective for early prediction
P. Cortez & A. Silva (2008)	Decision Tree, Random Forest, SVM, Neural Networks	Portuguese student dataset (grades, family background, social factors)	SVM & Neural Networks achieved best results	Previous grades and socio-demographic factors strongly affect performance
(MDPI Study, 2024)	SVM, Random Forest, Neural Networks, GAN, Regression Models	Academic + behavioral + demographic data	Accuracy improved by ~21% using regression models	Hybrid and regression models enhance prediction performance
(IntechOpen Study, ~2020)	Decision Tree, Random Forest, Naïve Bay	Educational datasets (attendance, grades, behavioral data)	ML models achieved good prediction accuracy	Data preprocessing and feature selection improve results

Scope

The scope defines the functional boundaries of this web-based platform, focusing on transforming raw educational data into predictive intelligence.

1. Data Analysis & Integration

The system processes multi-dimensional datasets to build a comprehensive student profile, including:

- **Attendance Records:** Monitoring consistency and engagement levels.
- **Internal Assessments:** Analyzing marks from quizzes and mid-terms.
- **Assignments:** Tracking submission timelines and completion quality.
- **Historical Data:** Incorporating past academic records to establish performance baselines.

2. Predictive ML Framework

The core engine utilizes supervised learning to forecast outcomes using:

- **Logistic Regression:** For binary classification (e.g., Pass/Fail probability).
- **Random Forest:** To handle complex, non-linear relationships between factors.
- **SVM (Support Vector Machine):** To precisely segment students in high-dimensional data spaces.

3. Classification & Risk Assessment

The system automatically categorizes students into actionable tiers:

- **High Performers:** Candidates for advanced tracks or mentorship.
- **Average:** Students showing steady progress but needing monitoring.
- **At-Risk:** Early flagging for students requiring immediate academic intervention.

4. Visualization & Interface

The platform provides a user-friendly, responsive web experience featuring:

- **Interactive Dashboards:** Real-time charts and heatmaps for faculty and admins.
- **Student/Faculty Portals:** Personalized views for tracking progress and managing alerts.
- **Automated Reporting:** Downloadable summaries for performance reviews.

Use Of Engineering

Use of Engineering Design Principles:

- Modular system design (data layer, application layer, presentation layer)
- Structured workflow including data collection, preprocessing, model training, and deployment
- User-centric design for easy interaction

Implementation of Algorithms:

- Machine learning algorithms such as Logistic Regression, Random Forest, and SVM are implemented
- Algorithms are trained and tested on student datasets to predict outcomes

Objectives

1. To Study the Existing Framework for Student Performance Evaluation

The first objective involves a comprehensive analysis of the current methods used by educational institutions to monitor student progress.

- **Identification of Data Silos:** Investigating how data (attendance, marks, assignments) is currently stored—often in fragmented manual registers or basic spreadsheets.
- **Analyzing the "Reactive" Model:** Evaluating the limitations of traditional systems, specifically the time lag between a student's academic decline and the institution's ability to detect it.
- **Feature Identification:** Determining which specific variables (e.g., mid-term scores vs. class participation) have the highest historical correlation with final success or failure.

2. To Develop a Predictive ML-Based Framework

The secondary objective is the technical core of the project: building an intelligent system that moves from observation to prediction.

- **Algorithm Integration:** Designing a multi-algorithm engine incorporating **Logistic Regression**, **Random Forest**, and **SVM** to handle different types of student data complexities.
- **Web-Based Platform Development:** Creating a centralized "Online" hub where faculty can upload data and instantly receive performance forecasts.
- **Risk Categorization:** Engineering a classification logic that automatically segments the student population into "High Performer," "Average," and "At-Risk" categories based on real-time data inputs.

3. To Confirm and Validate the Prepared Framework with Existing Standards

The final objective is to ensure that the new ML system is more effective and reliable than the traditional "manual guess" or "simple average" methods.

- **Comparative Analysis:** Running the ML models against historical "blind" data to see if the system could have predicted past failures more accurately than the teachers did at the time.
- **Accuracy Benchmarking:** Measuring the system's performance using engineering metrics like **Accuracy**, **Precision** (avoiding false alarms), and **Recall** (ensuring no at-risk student is missed).
- **Framework Alignment:** Ensuring the software architecture aligns with institutional requirements for data privacy (e.g., FERPA/GDPR) and usability, confirming it is a viable replacement for current outdated systems.

Methodology

The system follows a structured machine learning pipeline:

1)Data Collection:

The first step involves collecting relevant student data from various sources such as CSV files, academic databases, or institutional records. The dataset includes attributes like student marks, attendance, subject details, and other relevant academic information. This data serves as the foundation for analysis and model training.

2) Data Pre-Processing:

The collected data is often raw and may contain missing values, duplicate records, or inconsistencies. Therefore, preprocessing is performed to clean and prepare the data. This includes:

- Handling missing values
- Removing duplicates
- Normalizing and scaling data
- Encoding categorical variables

3. Data Architecture

In this stage, a structured data framework is designed to organize and store the processed data efficiently. The data is arranged into well-defined tables such as student, subject, and performance. Relationships between entities are established using database design concepts (ER diagram). This step ensures efficient data storage, retrieval, and scalability of the system.

4. Data Analysis and Modeling

After organizing the data, machine learning algorithms are applied to analyze patterns and predict student performance. Techniques such as:

- Logistic Regression
- Random Forest
- Support Vector Machine (SVM)

are used to train models on the dataset. Feature selection and engineering are also performed to improve prediction accuracy.

5. Model Evaluation

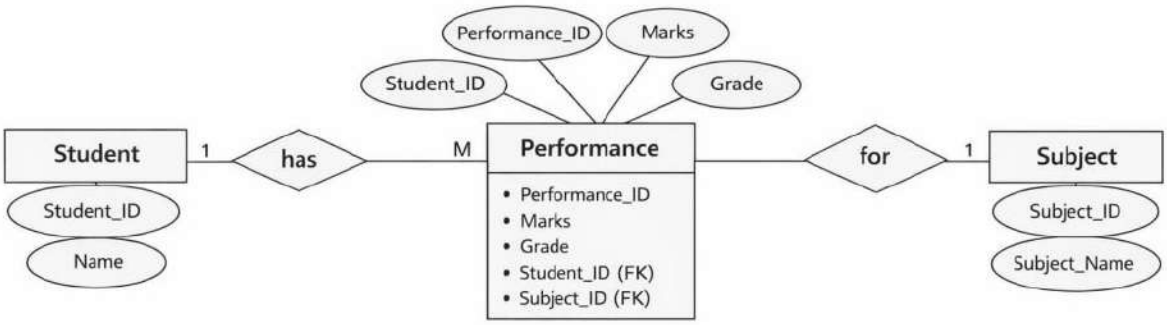
The performance of the trained models is evaluated using metrics such as:

- Accuracy
- Latency
- Precision
- Recall

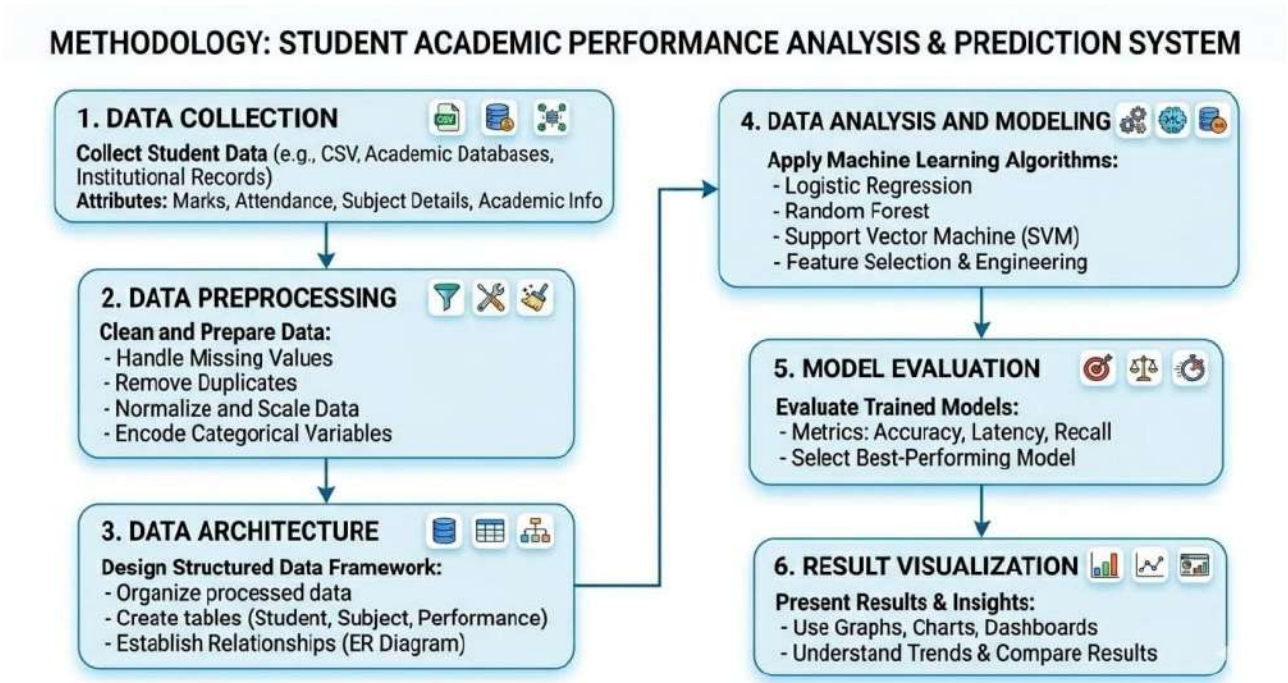
6. Result Visualization

The results obtained from the model are presented using graphs, charts, and dashboards. Visualization tools help in understanding performance trends and comparing results effectively.

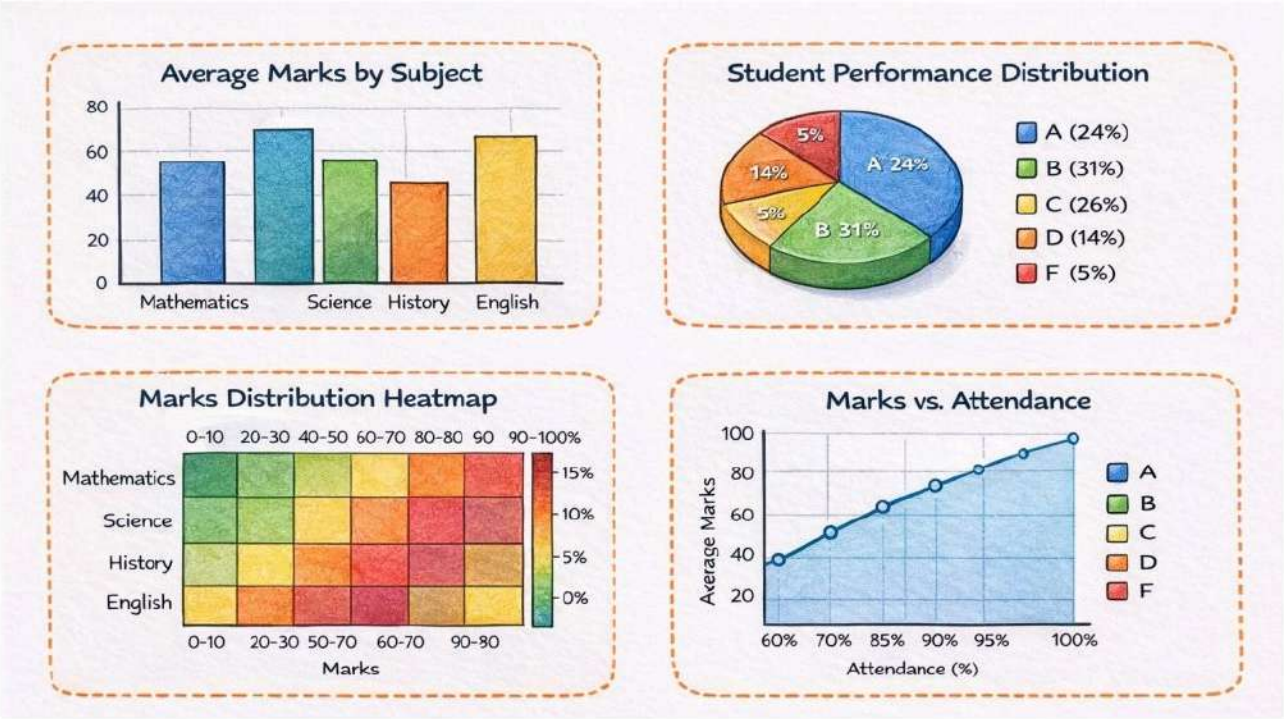
ER diagram



FlowChart:



Graphs


Average Marks by Subject
80
60
40
20
0
Mathematics
Science
History
English
Student Performance Distribution
A 24%
B 31%
14%
5%
5%
A (24%)
B (31%)
C (26%)
D (14%)
F (5%)
Marks Distribution Heatmap
0-10 20-30 40-50 60-70 80-80 90 90-100%
Mathematics
Science
History
English
15%
10%
5%
0%
0-10 20-30 50-70 60-70 90-80
Marks
Marks vs. Attendance
100
80
60
40
20
Average Marks
60% 70% 85% 90% 95% 100%
Attendance (%)
A
B
C
D
F

Tools and Technology

The development of the Online ML-Based Academic Performance Analyzer involves the use of various tools and technologies across different layers of the system:

1) Programming Language

? Python 3.14.3 - Used for implementing machine learning algorithms, backend logic, and data processing due to its simplicity and extensive ML ecosystem.

2) Machine Learning & Data Processing Libraries

? Scikit-learn – used for building and evaluating machine learning models such as Logistic Regression, Random Forest, and SVM

? Pandas – used for data manipulation, cleaning, and preprocessing

? NumPy – used for numerical computations and handling arrays

3) Data Visualization Tools

? Matplotlib – used for creating graphs such as bar charts and line plots

? Seaborn – used for advanced visualization like heatmaps and correlation plots

4) Frontend Technologies

? HTML, CSS, JavaScript - Used for designing responsive user interfaces and dashboards

5) Database

? MySQL - Used for storing student data, predictions, and system records

6) Development Tools

? Jupyter Notebook – used for data analysis, experimentation, and model training

? Visual Studio Code (VS Code) – used for application

Expected Outcomes

1. Predictive Accuracy

Accuracy is the primary metric for determining the reliability of the system's early warning signals. The goal is to minimize both "False Alarms" (flagging successful students as at-risk) and "Missed Cases" (failing to identify struggling students).

- **Model Performance Benchmarks:** The system aims for a classification accuracy of **85% to 95%** across the three core algorithms (**Logistic Regression, Random Forest, and SVM**).
- **Precision and Recall:** * **High Recall:** Ensuring that the system catches the maximum number of "At-Risk" students so that no one falls through the cracks.
    - **High Precision:** Ensuring that the interventions suggested by the system are credible and data-backed, maintaining faculty trust in the tool.
- **Feature Sensitivity:** The outcome is expected to show which factors (e.g., attendance vs. assignment scores) are the strongest predictors of success, allowing for "weighted" accuracy that reflects real-world academic patterns.

2. System Latency

Latency refers to the speed and responsiveness of the web-based platform. Since the system is designed for "Online" use by multiple faculty members and students simultaneously, high performance is critical.

- **Real-time Inference Speed:** Once data is uploaded, the ML models are expected to generate performance predictions in **under 2 seconds**. This ensures that educators receive "instant" feedback during data entry or bulk uploads.
- **Dashboard Responsiveness:** The visualization layer (charts, graphs, and heatmaps) should render within **1 to 3 seconds**, providing a fluid user experience even when processing large datasets of 1,000+ student records.
- **Data Processing Efficiency**: The preprocessing pipeline (cleaning and normalizing raw data) is optimized to handle high-volume inputs without "bottlenecking" the application server, ensuring the system remains stable during peak usage periods (e.g., mid-term grading weeks).

Applications

The Online ML-Based Academic Performance Analyzer has wide applications in the education sector and related domains:

- **Schools and Colleges**

To monitor student performance, identify weak students early, and improve academic outcomes

- **Universities and Educational Institutions**

For large-scale academic data analysis, performance tracking, and decision-making

- **Online Learning Platforms (LMS)**

To analyze student engagement and predict course completion or dropout risk

- **Coaching Institutes and Training Centers**

To evaluate student preparation levels and provide personalized guidance

- **Academic Counseling Systems**

To assist counselors in identifying at-risk students and suggesting improvement strategies

- **Educational Data Analytics Systems**

To generate insights from student data and improve teaching methodologies

- **Government and Scholarship Programs**

To identify deserving students and support data-driven policy decisions

- **Personalized Learning Systems**

To recommend study materials and learning paths based on student performance

Conclusion

The Online ML-Based Academic Performance Analyzer presents an effective solution to the limitations of traditional academic evaluation systems. By leveraging machine learning techniques, the system transforms raw student data into meaningful predictive insights. It enables early identification of at-risk students, allowing educators to take timely corrective actions and improve overall academic performance. The use of algorithms such as Logistic Regression, Random Forest, and Support Vector Machine ensures accurate and reliable predictions. The system also provides interactive dashboards and visualizations, making it easier for students, faculty, and administrators to understand performance trends and make data-driven decisions. Overall, this project demonstrates the practical application of machine learning in the education sector and highlights its potential to enhance learning outcomes, reduce manual effort, and support intelligent academic management systems.

Future Work

The proposed system can be further enhanced with advanced features and technologies to improve its functionality and scalability:

- Integration with Learning Management Systems (LMS) for automatic data collection
- Development of a mobile application for easy access and realtime notifications
- Real-time performance monitoring using APIs and live data integration
- Implementation of alert systems (email/SMS) for students, teachers, and parents
- Deployment on cloud platforms for scalability and multiinstitution usage

Guide Signature : _____

Date: 27/03/2026